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When Has a Bayesian Neural Network Sampled Enough? Adaptive Inference Time with Statistical Guarantees

Anytime-Valid Confidence Sequences Terminate Monte Carlo Sampling Per-Input When the Decision Is Statistically Resolved

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A fixed Monte Carlo sample budget in Bayesian neural network inference over-samples easy inputs and may under-sample hard ones, with no formal control on decision reliability. This paper terminates sampling per-input using *confidence sequences*—anytime-valid probability bounds—as soon as the relevant decision is statistically resolved, concentrating compute on ambiguous cases and providing a formal per-prediction guarantee.

AT A GLANCE

Problem: Fixed-budget BNN inference leaves finite-sample prediction error uncontrolled and wastes compute on easy inputs.

Why it matters: Reliable per-input guarantees are critical for high-stakes BNN deployment.

Method: Anytime-valid KL-mixture confidence sequences for argmax, distributional, and threshold BNN decisions.

Result: Adaptive sampling reduces latency vs. matched-reliability fixed budgets while providing a formal guarantee per prediction.

Evidence: Image classification benchmarks; comparison with fixed-S and heuristic early stopping.

The Big Picture

Bayesian neural networks approximate the posterior predictive distribution by averaging over S Monte Carlo weight samples:

$$\hat{p}(y \mid \mathbf{x}) = \frac{1}{S}\sum_{s=1}^{S} p(y \mid \mathbf{x}, \mathbf{w}^{(s)})$$

The choice of S is arbitrary, uniform across inputs, and provides no bound on the finite-sample error of the resulting prediction. For inputs far from the decision boundary, $S = 5$ is likely sufficient; for borderline cases, $S = 1000$ may still be insufficient. A method that terminates sampling when the decision is resolved—with a provable guarantee—would improve both reliability and efficiency simultaneously.

Why Existing Approaches Fall Short

Fixed-S inference ignores input difficulty, over-sampling easy inputs and providing no guarantee on hard ones.

Heuristic early stopping (stopping when the running mean stabilises) provides no formal guarantee and empirically fails near decision boundaries.

Conformal prediction provides marginal set-coverage guarantees over the test set, not per-input anytime guarantees tied to the sampling process.

Sequential hypothesis testing (SPRT) was designed for binary tests; multi-class prediction and distributional estimation require a different treatment.

Core Mechanism

A **confidence sequence** $\{C_s\}_{s=1}^{\infty}$ satisfies:

$$\Pr\big[\exists s \geq 1 : \theta \notin C_s\big] \leq \alpha$$

This anytime-valid property allows stopping at any data-dependent time τ without inflating the Type I error. Unlike classical confidence intervals (valid only at a fixed s), confidence sequences shrink as more samples arrive and remain valid throughout.

The paper instantiates confidence sequences for three BNN tasks: (1) argmax class identification, (2) full predictive distribution estimation, and (3) probability-threshold decisions.

Technical Formulation

Let $\hat{p}_s^c = \frac{1}{s}\sum_{j=1}^{s} p(y = c \mid \mathbf{x}, \mathbf{w}^{(j)})$. A KL-mixture confidence sequence gives an anytime interval (L_s^c, U_s^c):

$$\Pr\Big[\exists s \geq 1 : p^c \notin (L_s^c, U_s^c)\Big] \leq \alpha$$

The stopping rule for argmax identification is:

$$\tau^* = \min\Big\{s : U_s^{c^*} > \max_{c \neq c^*} U_s^c\Big\}$$

where $c^* = \arg\max_c \hat{p}_s^c$.

For distributional estimation with tolerance ε:

$$\tau^{\varepsilon} = \min\Big\{s : r_s \leq \varepsilon\Big\}, \quad r_s = O\left(\sqrt{\frac{\log s}{s}}\right)$$

For threshold decisions at level τ_{thr}:

$$\tau^{\tau_{\text{thr}}} = \min\Big\{s : L_s^{c^*} > \tau_{\text{thr}} \text{ or } U_s^{c^*} < \tau_{\text{thr}}\Big\}$$

Learning or Inference Procedure

At inference time, for each input $\mathbf{x}$:

1. Draw $\mathbf{w}^{(s)} \sim q(\mathbf{w} \mid \mathcal{D})$ and evaluate $p(y \mid \mathbf{x}, \mathbf{w}^{(s)})$.
2. Update running mean $\hat{\mathbf{p}}_s$ and CS bounds (L_s^c, U_s^c).
3. Check the stopping criterion for the chosen task.
4. If met, return the prediction with its guarantee; else draw $s+1$.

No training modifications are required; the method wraps any BNN.

What the Guarantee Says

For any data-dependent stopping time τ induced by the confidence sequence and any error level α, the decision at time τ has error probability at most α—regardless of the input distribution or Monte Carlo sample realisations.

Experimental Findings

- The method assigns **more samples to ambiguous inputs** (near decision boundaries) and fewer to easy inputs, naturally concentrating compute.
- **Overall inference latency is reduced** vs. a fixed-S budget that provides the same average reliability.
- CS bounds are **tight in practice**: average sample count closely tracks input difficulty measured by entropy.
- For threshold decisions, unnecessary high-sample expenditure on clearly low- or high-uncertainty inputs is avoided.

Ablations and Interpretation

Confidence level α: Stricter guarantees (lower α) require more samples per input; the relationship is approximately $O(\log(1/\alpha))$.

Posterior quality: Better-calibrated posteriors tighten CS bounds faster; an over-dispersed posterior still provides a valid guarantee but at greater sample cost.

Heuristic early stopping: Provides no guarantee and fails empirically on high-entropy inputs where the running mean converges slowly.

Dataset difficulty skew: Savings are largest on datasets with many easy and few hard examples—the common case in practice.

Reference

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